PAUL GERHART

mail@paul-gerhart.de | paul-gerhart.de | Privacy Enhancing Technologies, TU Wien

EDUCATION

09/2024 – present | **Doctoral Student in Cryptography**, TU Wien
Threshold Signatures, PoPRFs, Authentication Protocols (advisor: Dominique Schröder)

08/2022 – 08/2024 | **Doctoral Student in Cryptography**, FAU Erlangen-Nürnberg
Adaptor Signatures, Threshold Signatures (advisor: Dominique Schröder)

10/2020 – 07/2022 | **Master of Science in Computer Science**, FAU Erlangen-Nürnberg
Cryptography, Privacy, Cyber Security, and IT-Forensics (advisor: Dominique Schröder)

10/2017 – 09/2020 | **Bachelor of Science in Mathematics**, University of Bonn
Algebraic Topology (advisor: Wolfgang Lück)

EXPERIENCE

09/2024 – present | **University Research Assistant**, TU Wien
Privacy Enhancing Technologies

12/2022 – present | **Consultant and Software Developer**, Freelance
Cryptography, Privacy, IT-Forensics

08/2022 – 08/2024 | **University Research Assistant**, FAU Erlangen-Nürnberg
Applied Cryptography

10/2017 – 07/2022 | **Student Employee**, Richard Wenzel GmbH & Co.KG
Software Development

RESEARCH VISITS

06/2026 – 08/2026 | **Research Internship**, Prof. Emma Dauterman
Stanford University, Stanford, CA

08/2025 – 10/2025 | **Research Internship**, Prof. Thomas Ristenpart
Cornell Tech, New York, NY

08/2024 | **Research Visit**, Prof. Anna Lysyanskaya
Brown University, Providence, RI

08/2023 – 11/2023 | **Research Internship**, Prof. Jonathan Katz
University of Maryland, College Park, MD

HONORS & AWARDS

2025 | **Google PhD Fellowship** for Privacy, Safety, and Security
Google Research

ADVISED STUDENTS

2025 | **Jay Taylor** (Bachelor Thesis)
Delegated Fair Exchange via Threshold Wallets

2024 | **Fabian Jost** (Master Thesis)
Foundations of Blind Conditional Signatures

2023 | **Daniel Lang** (Bachelor Thesis)
Strong Unlinkability for Adaptor Signatures

2023 | **Ruben Baecker** (Master Thesis)
Phoenix Rise Once Again: How to Defeat the (PW-)Hero

2023 | **Julian Thomas** (Master Thesis)
Signature Security Revisited: One Dice to Rule Them All

TEACHING

Winter 25/26	**Privacy Enhancing Technologies (B.Sc.)**	Teaching assistant
Summer 25	**Advanced Privacy Enhancing Technologies (M.Sc.)**	Teaching assistant
Winter 24/25	**Privacy Enhancing Technologies (B.Sc.)**	Teaching assistant
Winter 23/24	**Introduction to Modern Cryptography**	Teaching assistant
Winter 22/23	**Introduction to Modern Cryptography**	Teaching assistant

COMMUNITY SERVICE

Program Committees

USENIX Security	2027
European Symposium on Research in Computer Security (ESORICS)	2026
SECRYPT	2025–2026
Crypto Valley Conference (CVC)	2025

External Reviews (Selected)

Advances in Cryptology (CRYPTO)	2023–2026
Advances in Cryptology (EUROCRYPT)	2023–2027
Advances in Cryptology (ASIACRYPT)	2022
ACM Conference on Computer and Communications Security (CCS)	2024–2026
Public Key Cryptography (PKC)	2023–2025
European Symposium on Research in Computer Security (ESORICS)	2022–2026
IEEE Computer Security Foundations Symposium (CSF)	2023
Financial Cryptography and Data Security (FC)	2024

PUBLICATIONS

13 peer-reviewed publications

- 8 publications at top cryptography conferences (CRYPTO, EUROCRYPT, ASIACRYPT)
- 2 publications at top security conferences (ACM CCS, NDSS)
- 2 publications at top privacy venues (PoPETS)

BitPriv: A Privacy-Preserving Protocol for DeFi Applications on Bitcoin — NDSS 2027
Ioannis Alexopoulos, Zeta Avarikioti, **Paul Gerhart**, Matteo Maffei, and Dominique Schröder

Fully Adaptive FROST with Identifiable Aborts from AOMDL — ASIACRYPT 2026
Ruben Baecker, **Paul Gerhart**, Davide Li Calsi, Luigi Russo, Dominique Schröder, and Arkady Yerukhimovich

Fair Distributed Exchange via Threshold Adaptor Signatures — ASIACRYPT 2026
Ruben Baecker, **Paul Gerhart**, Jonathan Katz, and Dominique Schröder

Universally Composable Adaptor Signatures — ACM CCS 2026
Paul Gerhart, Daniel Rausch, and Dominique Schröder

Fully-Adaptive Two-Round Threshold Schnorr Signatures from DDH — EUROCRYPT 2026
Paul Gerhart, Davide Li Calsi, Luigi Russo, and Dominique Schröder

How To Make Delegated Payments on Bitcoin: A Question for the AI Agentic Future — FC 2026
Jay Taylor, **Paul Gerhart**, and Sri Aravinda Krishnan Thyagarajan

Password-Hardened Encryption Revisited — ASIACRYPT 2025
Ruben Baecker, **Paul Gerhart**, and Dominique Schröder

Universally Composable Password-Hardened Encryption ASIACRYPT 2025
Behzad Abdolmaleki, Ruben Baecker, **Paul Gerhart**, Mike Graf, Mojtaba Khalili, Daniel Rausch, and Dominique Schröder

A Fully-Adaptive Threshold Partially-Oblivious PRF CRYPTO 2025
Ruben Baecker, **Paul Gerhart**, Daniel Rausch, and Dominique Schröder

SoK: Descriptive Statistics Under Local Differential Privacy PoPETS 2025
René Raab, Pascal Berrang, **Paul Gerhart**, and Dominique Schröder

Measuring Conditional Anonymity - A Global Study PoPETS 2024
Pascal Berrang, **Paul Gerhart**, and Dominique Schröder

Foundations of Adaptor Signatures EUROCRYPT 2024
Paul Gerhart, Dominique Schröder, Pratik Soni, and Sri Aravinda Krishnan Thyagarajan

Practical Schnorr Threshold Signatures without the Algebraic Group Model CRYPTO 2023
Hien Chu, **Paul Gerhart**, Tim Ruffing, and Dominique Schröder

A General Approach to Adaptor Signatures Master Thesis
advised by Dominique Schröder, Sri Aravinda Krishnan Thyagarajan, and Pratik Soni

Surgery and a Theorem of Lickorish Bachelor Thesis
advised by Wolfgang Lück and Christoph Winges